The Economics of Winning: Predicting Formula One Team Revenue and Championship Success from Financial & Performance Data

Course: Statistics & Machine Learning

SRH University of Applied Sciences Heidelberg

1. Introduction

This project applies a complete machine learning workflow to historical Formula One financial and performance data. Two supervised learning tasks were addressed: regression to predict next-season team revenue and classification to predict the Constructors' Champion. The objective was to investigate how financial indicators and racing performance can forecast future outcomes.

2. Dataset Description

The dataset was sourced from the Formula 1 Racing Industry dataset (2010–2026). Five relational tables containing financial, business, circuit, standings, and race result information were merged into a single team-season dataset. The final dataset contained 164 observations, 25 modelling features, and covered 17 seasons. Targets were next-season revenue (regression) and constructors' champion (classification).

3. Data Preprocessing

Structural missing values in the cost-cap variable before 2021 were filled with zero. Outliers were assessed using the IQR method and retained when they represented genuine Formula One spending. The datasets were merged by season. The categorical feature team_name was one-hot encoded using drop_first=True, leaving Alpine as the reference category. Numerical features were standardized using StandardScaler. Total revenue was removed from the predictors to prevent target leakage, and chronological train-test splitting was used to mimic real forecasting.

4. Feature Engineering and Selection

Engineered features included Sponsorship Efficiency, Budget Margin, Sponsorship Percentage, and Cost Cap Utilization. Feature importance was evaluated using correlation analysis, mutual information, ANOVA F-test, and Random Forest importance. Operating Budget, Sponsorship Revenue, Team, Season, and Cost Cap consistently ranked among the most informative features.

5. Statistical Analysis

Descriptive statistics, Shapiro–Wilk normality testing, Pearson and Spearman correlations, Welch's t-test, ANOVA, Kruskal–Wallis, and Chi-square analyses were performed. Visualizations included histograms, boxplots, scatter plots, violin plots, and heatmaps. Results showed a strong positive relationship between operating budget and prize money, while championship-winning teams generally had significantly larger budgets.

6. Model Training

Regression models evaluated included Naive, Linear Regression, Ridge Regression, Lasso Regression, and Random Forest. Lasso Regression was selected for its strong generalization performance. Classification models included Logistic Regression, Random Forest, and a Naive baseline. Logistic Regression was selected due to its performance and interpretability.

7. Model Evaluation

Regression was evaluated using R^2 and RMSE. The Lasso model achieved an R^2 of approximately 0.98 and an RMSE of approximately $23 million on holdout seasons. Classification was evaluated using ROC-AUC, Precision, Recall, F1-score, and champion prediction accuracy. The classifier correctly predicted 9 of 12 champion seasons with an ROC-AUC of approximately 0.91 and Precision, Recall, and F1 scores of approximately 0.75.

8. Interpretation

Operating Budget was the strongest predictor across multiple feature-selection methods, highlighting the close relationship between financial investment and competitive success. Both modelling approaches identified McLaren as the strongest candidate for future revenue and championship success. Limitations include the small dataset, class imbalance, reconstructed labels, and dependence on current Formula One regulations.

9. Conclusion

The project successfully demonstrated the full machine learning workflow: dataset preparation, preprocessing, feature engineering, statistical analysis, model training, evaluation, interpretation, and conclusion. The findings show that machine learning can effectively model Formula One financial and sporting performance while identifying areas for future improvement.

10. Team Contributions

The project was completed collaboratively, with responsibilities divided across the major stages of the machine learning workflow. Although each member focused on specific tasks, the team jointly reviewed the implementation, report, and presentation to ensure a shared understanding of the complete workflow.

Team Member	Responsibilities
Frederick Boafo	Data preprocessing, missing value handling, outlier analysis, one-hot encoding, feature scaling, and documentation of preprocessing workflow
Shashank Sridhar Kenkere	Multi-table merge (season join with f1_business.csv), encoding, engineered ratios, correlation/MI/ANOVA feature selection
Hrithik Wadile	Exploratory Data Analysis (EDA), descriptive statistics, correlation analysis, statistical hypothesis testing, and visualization generation.
Akshay Myakal	Revenue Predictive Modeling, Model Evaluation, Residual Analysis, Forecasting 2027 Revenue
Sahil Suryawanshi	Championship Classification Modeling, Model Evaluation, Confusion Matrices, Limitations

Collaboration Statement

While responsibilities were divided among team members, all major decisions regarding dataset preparation, preprocessing, feature engineering, model selection, evaluation, and interpretation were discussed collectively.